

Scam Signatures: Discovering Fraud Families in Online Job Postings

Project 1, Milestone 1: Proposal and Data Selection

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Topic

Scam Signatures: Discovering Fraud Families in Online Job Postings. This project applies unsupervised learning to real job postings to determine whether fraudulent listings share a small number of recognizable patterns and what language defines each one.

Business Problem

Job scams are a growing source of consumer harm. The Federal Trade Commission reported more than \$220 million in job scam losses in the first half of 2024 alone (Federal Trade Commission, 2024), and the Better Business Bureau found that employment scam reports doubled in 2025 (Better Business Bureau, 2026). Existing detection tools, including a classifier and a fine-tuned language model I built on this dataset in DSC630 and DSC670, answer only one question: is this posting fraudulent? A verdict without an explanation is hard for a job seeker to act on and hard for a job board's trust and safety team to audit. The business problem is that fraud detection tells users a posting is suspicious without telling them why. This project addresses it by asking whether fraudulent postings can be organized into a small, explainable set of scam families based on their language alone. The dataset is reused; the research questions, methods, code, and analysis are new to this course.

The specific research questions are:

1. When job postings are grouped by textual similarity alone, with the fraud label withheld, do fraudulent postings concentrate in a small number of clusters or scatter among legitimate postings? This will be measured by comparing each cluster's fraud rate to the dataset's base rate of roughly five percent and by scoring cluster agreement with the true labels.

2. How many distinct fraud signatures exist, and which words, phrases, and structural features (such as missing company profiles, compensation language, or contact instructions) distinguish each signature from legitimate postings?
3. Do job postings written in 2025 and 2026 fall into the signatures found in the 2012 to 2014 data, or do they form new groups the historical data does not capture?

My working hypotheses are that fraud will concentrate rather than scatter, that a minority of clusters will hold a majority of fraudulent postings, and that fraud signatures will be defined at least as much by what a posting omits as by what it says.

Datasets

The primary dataset is the Employment Scam Aegean Dataset (EMSCAD), published by the University of the Aegean's Laboratory of Information and Communication Systems Security and described in Vidros et al. (2017). It contains 17,880 real job postings collected through the Workable recruiting platform between 2012 and 2014, of which 866 (about 4.8 percent) were annotated as fraudulent by Workable's specialists. Each posting includes free-text fields (title, company profile, description, requirements, and benefits), categorical fields (employment type, required experience, required education, industry, and function), location and salary range, and three binary flags indicating whether the posting allowed telecommuting, displayed a company logo, or included screening questions. The text fields are the main input for this project; the categorical and binary fields will be used to describe and compare the clusters that emerge. I will work from the public Kaggle mirror of EMSCAD (Bansal, 2020), which contains the same 17,880 rows and 18 columns, and will remove exact duplicate postings and rows with empty description text before modeling.

The secondary dataset is a small evaluation set of job postings from 2025 and 2026, used to test whether the signatures found in the 2012 to 2014 data still describe current scams. My first choice is real postings and recruitment messages reproduced in credible public sources, including the Federal Trade Commission’s data spotlights, individual public reports on the Better Business Bureau’s Scam Tracker, and news coverage that quotes scam messages verbatim. Each item will be logged with its source, date, and the reason it was labeled a scam, and any names, phone numbers, or other identifying details will be removed before use. Newer job posting datasets on public repositories will be considered only if their origin and labeling method can be verified, since several do not document either. As a fallback, I retain a set of synthetic postings I wrote during the DSC670 project using terminology from current fraud reports; because I authored those with knowledge of what scams look like, they will be clearly labeled as synthetic and used only to supplement, not replace, sourced examples. Both sourced and synthetic postings are used for evaluation only, never for fitting.

Methods

The core pipeline has four stages. First, I will assemble each posting’s text fields into a single document, strip residual HTML and boilerplate, and convert each document to a dense vector using a pretrained sentence embedding model such as all-MiniLM-L6-v2 from the Sentence-BERT family (Reimers & Gurevych, 2019). Sentence embeddings capture meaning rather than exact wording, which matters because scam postings often paraphrase the same pitch. Second, I will reduce the embeddings with UMAP (McInnes et al., 2018), which preserves local structure well enough for density-based clustering and produces a two-dimensional map for visualization. Third, I will cluster the reduced embeddings with HDBSCAN (McInnes et al., 2017) as the primary method, because it does not require

choosing the number of clusters in advance and can label ambiguous postings as noise rather than forcing them into a group. K-means with silhouette-based selection of k will be run as a comparison, and a TF-IDF plus K-means baseline will show how much the embeddings add over word counts. Fourth, I will interpret the clusters with class-based TF-IDF to surface the distinguishing terms of each group, latent Dirichlet allocation (Blei et al., 2003) to summarize the themes within fraud-heavy clusters, and comparisons of the structural fields (for example, the share of postings in a cluster with no company profile) to describe each fraud signature in plain language.

Validation works in two directions. Internally, silhouette scores (Rousseeuw, 1987) and cluster stability across random seeds will show whether the groupings are real rather than artifacts of one run. Externally, the fraud label, withheld during clustering, will be revealed afterward to compute each cluster's fraud rate, cluster purity, the adjusted Rand index (Hubert & Arabie, 1985), and normalized mutual information against the true labels. For the third research question, the 2025 and 2026 postings will be embedded with the same model and assigned to the nearest existing cluster, with their distance and outlier scores showing whether they fit the historical signatures or sit apart from them. Alternatives considered include DBSCAN applied directly to the high-dimensional embeddings, which tends to collapse into one large cluster and noise, and BERTopic (Grootendorst, 2022), which packages a similar embed, reduce, cluster, and describe pipeline and may be used if it proves more practical than assembling the stages by hand. The project can adapt as the data dictates.

Ethical Considerations

The following concerns apply to this topic and to analyzing this data. Each is rated for how likely it is to occur and how serious its impact could be, with an initial idea for mitigation that will be developed fully in the white paper’s Ethical Assessment.

Concern	Likelihood	Impact	Initial mitigation
Harm to legitimate employers. A cluster labeled as a fraud signature will also contain legitimate postings, and public use of the results could stigmatize honest employers whose listings resemble a scam pattern, such as small businesses with no company profile or remote data-entry roles. No cluster will be purely fraudulent, and the harm compounds if signatures are used to filter listings automatically.	High	Moderate to high	Report fraud rates per cluster rather than labeling clusters as fraud; present signatures as reasons for closer review rather than verdicts; never publish employer names from the data.

Concern	Likelihood	Impact	Initial mitigation
Label validity and dataset age. EMSCAD's fraud labels were assigned by Workable staff between 2012 and 2014, and the annotation process is described only in general terms. Some labels may be wrong, and some scam patterns from that period may no longer exist, so findings could describe historical rather than current fraud.	High	Moderate	Treat the labels as a reference point rather than ground truth; report agreement metrics with that caveat; use the 2025 and 2026 evaluation set to test whether signatures still apply.
Privacy of the people in the data. Postings contain company names, locations, and occasionally contact details, and sourced modern examples may come from victims' public reports; republishing identifying details could expose an employer to harassment or re-expose a victim.	Moderate	Moderate	Use only the text needed for analysis; remove names, phone numbers, emails, and URLs from any quoted example; cite sources for modern examples without reproducing personal narratives.

Concern	Likelihood	Impact	Initial mitigation
Potential for misuse. A clear description of what makes a scam posting recognizable is also a guide to writing one that avoids recognition, although the signatures describe patterns scammers already use.	Low	Moderate	Describe signatures at the level of themes and structural gaps rather than publishing word lists that could be inverted into a template; frame results for job seekers and trust and safety teams.
Fairness across industries and regions. If certain industries or countries are overrepresented among fraudulent postings, the signatures could encode that skew and disadvantage legitimate postings from those groups.	Moderate	Moderate	Report the industry and location composition of each cluster alongside its fraud rate; flag any signature whose distinguishing features are demographic rather than behavioral.
Researcher bias in the evaluation set. The synthetic postings I wrote reflect my own expectations of what scams look like, so strong results on them would partly confirm my assumptions rather than the model.	High if used alone	Low to moderate	Prioritize sourced real examples; label synthetic postings as such wherever they appear; report results on the two subsets separately.

Concern	Likelihood	Impact	Initial mitigation
Data sourcing and terms of use. EMSCAD is published for research use, but modern examples will come from several public sources with different terms.	Low	Low	Use the Kaggle mirror under its stated license; collect modern examples by hand from public pages rather than by scraping; respect each source's terms; document every item's origin.

Challenges/Issues

The most likely technical challenge is that clustering high-dimensional text does not always produce clean groups. HDBSCAN may label a large share of postings as noise, or the clusters may organize by industry and job function rather than by fraud, which would answer the first research question in the negative but still be a valid finding. Parameter choices for UMAP and HDBSCAN can change the results, so I will need to test stability across settings and seeds rather than reporting one configuration. The dataset's imbalance, with fewer than 900 fraudulent postings among nearly 18,000, means fraud-heavy clusters may be small and easy to overinterpret. Interpreting clusters in plain language is a challenge of its own; distinguishing terms and topics do not always translate into a signature a job seeker would recognize, and I will need to read postings from each cluster rather than rely on statistics alone. Assembling a credible modern evaluation set within the project timeline is a practical risk, since sourced examples must be located, verified, and cleaned by hand. Finally, four weeks is a short window for an unsupervised project with three research questions, so I plan to complete the core pipeline and the first two questions before investing in the third.

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